

Danny Yu-Chung Wang

Senior Data Scientist | Applied AI | PhD Candidate

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Born: 24/03/1990 | Nationality: Taiwan (Norway Permanent Residence) | Gender: Male



Summary

Senior Data Scientist with 7+ years of industry experience delivering AI-driven solutions in manufacturing and real-world systems. Award-winning project leader. PhD candidate focusing on customer behavior prediction, explainable AI, and practical, high-impact AI products.

Technical Skills

Visualization & LLM

PowerBI / Tableau / Streamlit | LLM application | RAG (retrieval, prompting, evaluation)

Applied ML & Modeling

Time-series & sequential modeling | Computer Vision (YOLO, ViT, SAM) | Explainable AI | Databricks

Python & Data Foundations

Python (Pandas, NumPy) | SQL & databases (e.g., PostgreSQL) | data wrangling & experimentation

Work Experience

PhD Research Fellow (AI/ML/Data Mining)

May 2023 – Present

Western Norway University of Applied Sciences (HVL) — Publicly funded research Bergen, Norway

- **Research focus:** Applied AI for **customer behavior prediction** and **explainable machine learning (XAI)**.
 - End-to-end ownership: problem formulation, modeling, evaluation, and communication.
 - Delivered peer-reviewed publications, research prototypes, and visualization tools.
- **Teaching (Master level):**
 - ADA511 *Datavitenskap og KI-prototyping* —lectures on Machine Learning and RAG.
 - Course materials: [Click here](#).
- **Honors & Funding:**
 - Visiting Scholar, R.C. Trustworthy Data Science and Security, TU Dortmund University (Germany).
 - Mobility Grant, HVL, Apr–Jul 2026.
 - Best Poster Award, PhD Seminar (2025, Voss).
- **Thesis:** *Customer Behavior Prediction and Model Explainability*.
- **Supervisors:** Prof. Lars Arne Jordanger (lars.arne.jordanger@hvl.no); Prof. Mojtaba Yousefi (mojtaba.yousefi@hvl.no).
- **Status:** All program requirements completed; on track for thesis submission in 2026.

Senior Data Scientist (Sr. Data Analytics Engineer), Manufacturing Technology & Engineering Asia

Sep 2019 – Mar 2023

Corning Inc. (Fortune 500; world-leading materials science company) Taichung, Taiwan

- **Key projects (Algorithm Owner / Project Leader):**
 - **Predictive Maintenance (PdM) for High Voltage Power Systems** Algorithm developer and data pipeline owner; delivered **\$2M CAPEX** impact across **6 factories** in Taiwan, Japan, and China.
 - **Glass product traceability & quality prediction systems** Project Leader for Sheet Resume (Sun Moon Lake) and quality prediction platforms; achieved **\$500K CAPEX** impact across factories in Taiwan, Korea, and China.
 - **Automotive glass coating process optimization** Designed data-driven algorithms for process fine-tuning, improving production yield by **20%**.
- **Collaboration & scale:**
 - Worked with cross-functional engineering, manufacturing, and IT teams across **25+ plants** in Asia.
- **Honors & recognition:**
 -  **Global Manufacturing Leadership Award** (2022) AI/Machine Learning, Project Leader; Manufacturing Leadership Council (MLC).
 - **Other awards:** Function Excellence Award (2021); Digital Summit Best Performance Award (2020); Best Poster Award (Corning Technical Conference).
- Award page: [Click here](#) | Press coverage: [Click here](#)

Senior System Engineer

Sep 2016 – Sep 2019

Powertech Technology Inc. – Top-5 global semiconductor assembly & test company Hsinchu, Taiwan

- Developed and maintained engineering systems for large-scale semiconductor manufacturing.
- Supported data-driven solutions for process optimization and yield improvement.
- **Award:** Outstanding Staff Award (2018).

System Engineer

Nov 2015 – Sep 2016

EPISTAR Inc. Hsinchu, Taiwan

- Production-level engineering tasks in LED manufacturing environments; Advanced Process Control system development.

Education

MSc, Industrial Engineering & Engineering Management (GPA: 3.82/4.3)

Sep 2012 – Jun 2014
National Tsing Hua University Hsinchu, Taiwan

- Research on data-driven decision support and energy data modeling.

BSc, Applied Mathematics Sep 2008 – Jun 2012

National Cheng Chi University Taipei, Taiwan

- Foundation in applied mathematics, statistics, and analytical modeling.

Selected Publications

- Wang, D. Y. C., Jordanger, L. A., & Lin, J. C. W. (2025). Visualizing Clickstream Prediction Logic and Utility. In Proceedings of the 27th European Conference on Artificial Intelligence (ECAI 2025).
- Wang, D. Y. C., Jordanger, L. A., & Lin, J. C. W. (2025). Explainability of highly associated fuzzy churn patterns in binary classification. Expert Systems, 42(7), e70066.
- Wang, D. Y. C., Jordanger, L. A., & Lin, J. C. W. (2024, December). A utility-mining-driven active learning approach for analyzing clickstream sequences. In 2024 IEEE International Conference on Big Data (BigData) (pp. 8363-8369).
- Wang, D. Y. C., Jordanger, L. A., & Lin, J. C. W. (2023, December). Explainability of leverage points exploration for customer churn prediction. In 2023 IEEE International Conference on Big Data (BigData) (pp. 5997-6004).
- Wang, D. Y. C., & Lin, J. C. W. (2023). Artificial Intelligence and Optimization Strategies in Industrial IoT Applications. In Industry 4.0 and Healthcare: Impact of Artificial Intelligence (pp. 223-251). Springer.

Side & Personal Projects

SankeyX (XAI Visualization for Sequence Models) 2025

Research / Open-source project

- Proposed SankeyX, a novel visualization method for explaining sequence models with uncertainty-aware views, addressing common limitations of prior XAI plots (e.g., cluttered flows and hard-to-compare attributions).
- Built SankeyX+, an LLM-assisted variant that supports interactive interpretation on top of dynamic-state annotations. Demos: GitHub repo: (Click here) | interactive demo: (Click here) | YouTube: (Click here)

LEGO Piece Recognition System 2025–Present
Personal Project

- Designed and trained a computer vision model for LEGO piece recognition, inspired by hands-on building with my child. We evaluated limitations of existing tools and explored inference and deployment constraints for consumer-facing AI, aiming to open-source a state-of-the-art solution.

Decision Support System for House Bidding 2025–Present
Personal Project (funding application in progress: 250K NOK)

- Developing a decision-support tool; currently applying for external funding.
- Open-source project: Interaktivt kart over barneskoler i Bergen. Project link: Click here.

Languages

English	Professional working proficiency
Norwegian	Intermediate
Chinese	Native

References

- Prof. Lars Arne Jordanger (HVL Supervisor) — lars.arne.jordanger@hvl.no
- Prof. Mojtaba Yousefi (HVL Supervisor) — mo-jtaba.yousefi@hvl.no
- Randy Kang (Corning Inc.) — kangs6@corning.com